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Roll No

EC-705-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

VLSI Design

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What do you mean by device count? Discuss the complexity of VLSI circuit.
b) Give the types of major processes used in IC fabrication.
2. Discuss the design and applications of bit serial processing elements.
3. a) Discuss the process of photolithography. Explain the various types of etching process.
b) Draw and explain the Y chart of device modeling.
4. Draw a relation for the subthreshold operation, how we can implement on short channel device.
5. Derive a model for estimating temperature dependence of BJT.
6. Discuss the high frequency BJT model along with governing equations.

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7. Discuss the design and applications of bit serial processing elements.
8. Write short notes: (any four)
 - a) Latch up
 - b) Micro coded controllers
 - c) BJT Noise Model
 - d) Device Modelling
 - e) Twin Tub Process
